GENERAL SCIENCE: BIOLOGY AND PHYSIOLOGY

CELL AND ORGANELLE MAP → MEMBRANE TRANSPORT DECISION TREE → CELL DIVISION COMPARISON → GENE-EXPRESSION RAIL → NUTRITION AND DIGESTION ROUTE → TRANSPORT RESPIRATION EXCRETION LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 CELL AND ORGANELLE MAP

CELL AND ORGANELLE MAP · NO MEMBRANE-BOUND NUCLEUS OR ORGANELLES · NUCLEUS + COMPARTMENTAL ORGANELLES · ER + GOLGI · PROCESSING AND MOVEMENT · CELLULOSE WALL + CHLOROPLAST + LARGE VACUOLE

PROKARYOTE → no membrane-bound nucleus or organelles

EUKARYOTE → nucleus + compartmental organelles → RIBOSOME → protein → MITOCHONDRION → ATP

CONCEPT / ARCHITECTURE

- PROKARYOTE → no membrane-bound nucleus or organelles
- EUKARYOTE → nucleus + compartmental organelles

MECHANISM / APPLICATION

- RIBOSOME → protein
- MITOCHONDRION → ATP

READINESS / STATUS LIMIT

- ER + GOLGI → processing and movement
- PLANT → cellulose wall + chloroplast + large vacuole

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLANT → cellulose wall + chloroplast + large vacuole.

01

STAGE 01 MEMBRANE TRANSPORT DECISION TREE

MEMBRANE TRANSPORT DECISION TREE · SOLUTE DOWN GRADIENT · WATER + SELECTIVE MEMBRANE · AGAINST GRADIENT + ENERGY · ACTIVE TRANSPORT · EXTERNAL TONICITY · CELL WATER RESPONSE · MOVEMENT DEPENDS ON MEMBRANE AND GRADIENT

SOLUTE DOWN GRADIENT → diffusion

WATER + SELECTIVE MEMBRANE → osmosis → AGAINST GRADIENT + ENERGY → active transport → EXTERNAL TONICITY → cell water response

CONCEPT / ARCHITECTURE

- SOLUTE DOWN GRADIENT → diffusion
- WATER + SELECTIVE MEMBRANE → osmosis

MECHANISM / APPLICATION

- AGAINST GRADIENT + ENERGY → active transport
- EXTERNAL TONICITY → cell water response

READINESS / STATUS LIMIT

- TRAP → movement depends on membrane and gradient
- SOLUTE DOWN GRADIENT → diffusion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → movement depends on membrane and gradient.

02

STAGE 02 CELL DIVISION COMPARISON

CELL DIVISION COMPARISON · ONE DIVISION · TWO SIMILAR CELLS · GROWTH AND REPAIR · TWO DIVISIONS · FOUR HAPLOID PRODUCTS · GAMETES + VARIATION · CROSSING OVER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
MITOSIS → one division → two similar cells	MEIOSIS → two divisions → four haploid products	CROSSING OVER != gene editing
ROLE → growth and repair	ROLE → gametes + variation	MITOSIS → one division → two similar cells

SYSTEM / DEVICE / INSTITUTION

- MITOSIS → one division → two similar cells
- ROLE → growth and repair

DECISIVE TECHNICAL DISTINCTION

- MEIOSIS → two divisions → four haploid products
- ROLE → gametes + variation

STATUS / UNIT / EVIDENCE LIMIT

- CROSSING OVER != gene editing
- MITOSIS → one division → two similar cells

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEIOSIS → two divisions → four haploid products.

03

STAGE 03 GENE-EXPRESSION RAIL

GENE-EXPRESSION RAIL · DNA --REPLICATION--> DNA · DNA --TRANSCRIPTION--> MRNA MRNA + TRNA + RIBOSOME --TRANSLATION--> · POST-TRANSCRIPTIONAL REGULATION · CELL STRUCTURE

CONCEPT / ARCHITECTURE

- DNA --replication--> DNA
- DNA --transcription--> mRNA mRNA + tRNA + RIBOSOME --translation--> protein microRNA → post-transcriptional regulation

MECHANISM / APPLICATION

- PROTEIN → cell structure / enzyme / receptor / signal
- DNA --replication--> DNA

READINESS / STATUS LIMIT

- DNA --transcription--> mRNA mRNA + tRNA + RIBOSOME --translation--> protein microRNA → post-transcriptional regulation
- PROTEIN → cell structure / enzyme / receptor / signal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROTEIN → cell structure / enzyme / receptor / signal.

04

STAGE 04 NUTRITION AND DIGESTION ROUTE

NUTRITION AND DIGESTION ROUTE · ACID + PEPSIN · FAT EMULSIFICATION · PANCREAS + INTESTINE · DIGESTION COMPLETION · SMALL INTESTINE · PRINCIPAL ABSORPTION

CONCEPT / ARCHITECTURE

- MOUTH → amylase
- STOMACH → acid + pepsin

MECHANISM / APPLICATION

- LIVER / BILE → fat emulsification
- PANCREAS + INTESTINE → digestion completion

READINESS / STATUS LIMIT

- SMALL INTESTINE → principal absorption
- MOUTH → amylase

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PANCREAS + INTESTINE → digestion completion; SMALL INTESTINE → principal absorption.

05

STAGE 05 TRANSPORT RESPIRATION EXCRETION LOOP

TRANSPORT RESPIRATION EXCRETION LOOP · ALVEOLAR GAS EXCHANGE · GASES + NUTRIENTS + HORMONES + WASTES · PULMONARY + SYSTEMIC CIRCULATION · ATP-PRODUCING RESPIRATION · FILTRATION + REABSORPTION + SECRETION

LUNGS → alveolar gas exchange → BLOOD → gases + nutrients + hormones + wastes → HEART → pulmonary + systemic circulation → CELLS → ATP-producing respiration → NEPHRON → filtration + reabsorption + secretion

CONCEPT / ARCHITECTURE

- LUNGS → alveolar gas exchange
- BLOOD → gases + nutrients + hormones + wastes

MECHANISM / APPLICATION

- HEART → pulmonary + systemic circulation
- CELLS → ATP-producing respiration

READINESS / STATUS LIMIT

- NEPHRON → filtration + reabsorption + secretion
- LUNGS → alveolar gas exchange

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOOD → gases + nutrients + hormones + wastes.

06

STAGE 06 CONTROL AND FEEDBACK LADDER

CONTROL AND FEEDBACK LADDER · CHANGE DETECTED · RAPID WIRED RESPONSE · BLOOD-BORNE HORMONE RESPONSE · PHYSIOLOGICAL ADJUSTMENT · NEGATIVE FEEDBACK · RESTORES REGULATED RANGE

RECEPTOR → change detected

NERVOUS → rapid wired response → ENDOCRINE → blood-borne hormone response → TARGET → physiological adjustment

STARTING CONDITION / INPUT

- RECEPTOR → change detected
- NERVOUS → rapid wired response

SCIENTIFIC OR ENGINEERING MECHANISM

- ENDOCRINE → blood-borne hormone response
- TARGET → physiological adjustment

OUTPUT / FAILURE BOUNDARY

- NEGATIVE FEEDBACK → restores regulated range
- RECEPTOR → change detected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TARGET → physiological adjustment; NEGATIVE FEEDBACK → restores regulated range.

07

STAGE 07 IMMUNITY ARCHITECTURE

IMMUNITY ARCHITECTURE · BARRIERS + PHAGOCYTES + COMPLEMENT · B CELL · T CELL · REGULATORY ROLES · REGULATORY T CELL + FOXP3 · PERIPHERAL TOLERANCE · SELF-REACTIVITY CONTROL

CONCEPT / ARCHITECTURE

- BARRIERS + PHAGOCYTES + COMPLEMENT → innate
- B CELL → antibody / memory

MECHANISM / APPLICATION

- T CELL → helper / cytotoxic / regulatory roles
- REGULATORY T CELL + Foxp3 → peripheral tolerance

READINESS / STATUS LIMIT

- SELF-REACTIVITY CONTROL != no immune response
- BARRIERS + PHAGOCYTES + COMPLEMENT → innate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY T CELL + Foxp3 → peripheral tolerance.

08

STAGE 08 DISEASE TRANSMISSION GRID

DISEASE TRANSMISSION GRID · CARRIES PATHOGEN, IS NOT DISEASE · ANIMAL-HUMAN INTERFACE · MECHANISM-SPECIFIC LAYERS

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
VECTOR → carries pathogen, is not disease	PREVENTION → mechanism-specific layers	ZOONOSIS → animal-human interface
ZOONOSIS → animal-human interface	VECTOR → carries pathogen, is not disease	PREVENTION → mechanism-specific layers

CONCEPT / ARCHITECTURE

- VECTOR → carries pathogen, is not disease
- ZOONOSIS → animal-human interface

MECHANISM / APPLICATION

- PREVENTION → mechanism-specific layers
- VECTOR → carries pathogen, is not disease

READINESS / STATUS LIMIT

- ZOONOSIS → animal-human interface
- PREVENTION → mechanism-specific layers

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AIR | WATER/FOOD | BLOOD | CONTACT | VECTOR.

09

STAGE 09 MICROBIAL ECOLOGY AND RESISTANCE

MICROBIAL ECOLOGY AND RESISTANCE · ANTIMICROBIAL PRESSURE · RESISTANT MICROBE · SURVIVAL + SPREAD · MATRIX-PROTECTED COMMUNITY · CONTEXT-DEPENDENT ROLES · EXTREMOPHILE LABEL · TOLERANCE, NOT UNIVERSAL SAFETY

ANTIMICROBIAL PRESSURE → selection

RESISTANT MICROBE → survival + spread → BIOFILM → matrix-protected community → MICROBIOME / PROBIOTIC → context-dependent roles

CONCEPT / ARCHITECTURE

- ANTIMICROBIAL PRESSURE → selection
- RESISTANT MICROBE → survival + spread

MECHANISM / APPLICATION

- BIOFILM → matrix-protected community
- MICROBIOME / PROBIOTIC → context-dependent roles

READINESS / STATUS LIMIT

- EXTREMOPHILE LABEL → tolerance, not universal safety
- ANTIMICROBIAL PRESSURE → selection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MICROBIOME / PROBIOTIC → context-dependent roles; EXTREMOPHILE LABEL → tolerance, not universal safety.

10

STAGE 10 PLANT FUNCTION FLOW

PLANT FUNCTION FLOW · WATER + MINERAL UPTAKE · UPWARD TRANSPORT · STOMATA + PHOTOSYNTHESIS + TRANSPIRATION · FOOD TRANSLOCATION · DIFFERENT CARBON-FIXATION STRATEGIES

ROOT → water + mineral uptake → XYLEM → upward transport → LEAF → stomata + photosynthesis + transpiration → PHLOEM → food translocation → CAM → different carbon-fixation strategies

STARTING CONDITION / INPUT

- ROOT → water + mineral uptake
- XYLEM → upward transport

SCIENTIFIC OR ENGINEERING MECHANISM

- LEAF → stomata + photosynthesis + transpiration
- PHLOEM → food translocation

OUTPUT / FAILURE BOUNDARY

- CAM → different carbon-fixation strategies
- ROOT → water + mineral uptake

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: C3 | C4 | CAM → different carbon-fixation strategies.

11

STAGE 11 PYQ AND EVIDENCE-STATUS RAIL

PYQ AND EVIDENCE-STATUS RAIL · PLANT FORM · SPECIES PROPERTY · EXACT CLASSIFICATION · MICROBIAL METABOLISM · PROGRAMME DELIVERY · WHO VALIDATION · PATHOGEN ERADICATION

TAXON / PLANT FORM / SPECIES PROPERTY → exact classification → CELLULOSE → cellulase → sugars → microbial metabolism → RESEARCH → SURVEILLANCE → PROGRAMME DELIVERY → REGULATION → WHO VALIDATION != pathogen eradication → MECHANISM / ASSOCIATION / APPROVAL / OUTCOME → separate rungs

CONCEPT / ARCHITECTURE

- TAXON / PLANT FORM / SPECIES PROPERTY → exact classification
- CELLULOSE → cellulase → sugars → microbial metabolism

MECHANISM / APPLICATION

- RESEARCH → SURVEILLANCE → PROGRAMME DELIVERY → REGULATION
- WHO VALIDATION != pathogen eradication

READINESS / STATUS LIMIT

- MECHANISM / ASSOCIATION / APPROVAL / OUTCOME → separate rungs
- TAXON / PLANT FORM / SPECIES PROPERTY → exact classification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MECHANISM / ASSOCIATION / APPROVAL / OUTCOME → separate rungs.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · BIOTECHNOLOGY POLICY AND PROGRAMME INSTITUTION LINKING BASIC BIOLOGY TO · MOLECULAR BIOLOGY, GENOMICS AND SYSTEMS-BIOLOGY RESEARCH ECOSYSTEMS: · DISEASE-SURVEILLANCE AND PUBLIC-HEALTH INSTITUTIONS · WHERE PHYSIOLOGY, MICROBIOLOGY AND IMMUNOLOGY BECOME POLICY ACTION. · SEPARATE THE FUNCTIONS PRECISELY · ICMR PERFORMS AND FUNDS BIOMEDICAL RESEARCH · NCDC AND THE IDSP/IHIP NETWORK UNDER MOHFW RUN OPERATIONAL

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- DBT: biotechnology policy and programme institution linking basic biology to applied missions.
- CSIR-CCMB / CSIR-IGIB: molecular biology, genomics and systems-biology research ecosystems.
- ICMR / disease-surveillance and public-health institutions: where physiology, microbiology and immunology become policy action.
- CAUTION: Separate the functions precisely: ICMR performs and funds biomedical research

READINESS, UNIT OR STATUS LIMIT

- NCDC and the IDSP/IHIP network under MoHFW run operational surveillance and outbreak response
- State health departments deliver programmes
- CDSCO grants regulatory approvals.
- Research capacity, surveillance capacity and regulatory capacity fail in different ways and require different remedies.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- WHO-linked surveillance and immunization reporting frameworks: relevant because public-health biology is continuously monitored, not studied once and forgotten.
- CAUTION: Applied extensions sit in 13 Biotechnology-Fundamentals-and-DBT-Missions.md, 14 Genetic-Engineering-GM-Crops-and-CRISPR.md and 15 Vaccines-Monoclonal-Antibodies-and-Biopharma.md.
- Genetics and cell biology underpin newborn screening, cancer diagnostics, pathogen surveillance and precision medicine.
- Physiology underlies nutrition programmes, anemia policy, non-communicable disease prevention and occupational health.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels